



Apple’s photographs, in matching rounded frames. The pair is the hero on apple.com/airpods-pro. The open case is the Apple Store gallery. The close-up is the heart-rate frame on the same product page. The numbered plates that follow mark only the part that frame actually shows.

AirPods Pro 3

A research note on one pair of earbuds. Listening modes, the decibel figures Apple actually prints, a laboratory frequency curve, every stem gesture, what each microphone is for, and

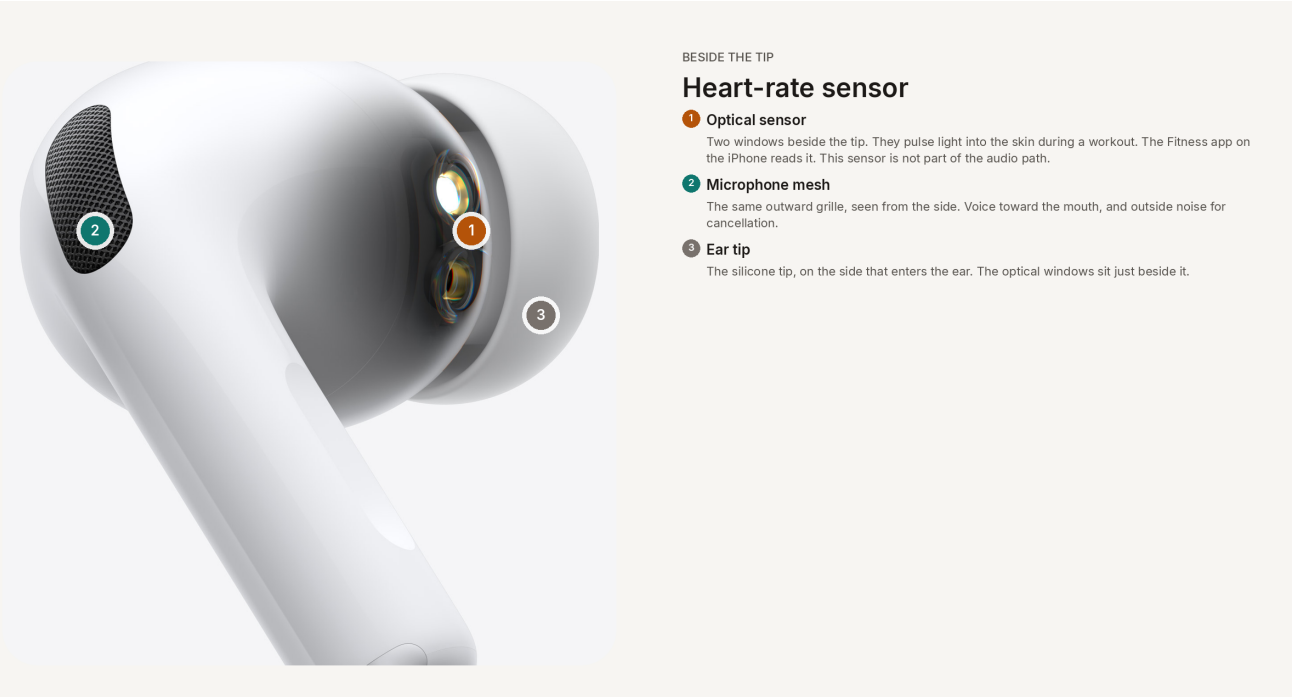
which speech features turn sound into text.

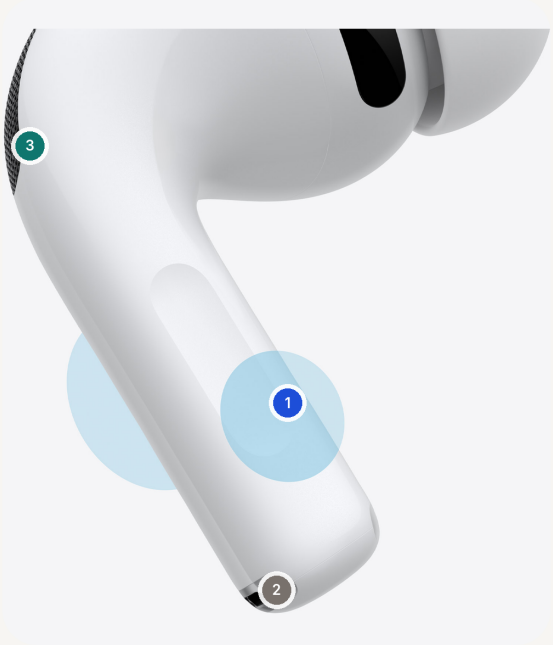
Announced 9 September 2025. In stores 19 September 2025. 249 US dollars. Apple H2 in each earbud, Bluetooth 5.3, IP57 on the earbuds and the case. Each earbud is 30.9 × 19.2 × 27.0 mm and weighs 5.55 g. Five foam-infused tips, XXS through L. Nothing in this note was measured on hardware in this project. Where Apple does not publish a number, the number is left out.

What each part is

Each plate is one of Apple’s own section photographs. The number on the photograph matches the note beside it. Where Apple shows a mesh and does not name it, the note says so.







APPLE'S STEM PHOTOGRAPH

The control surface

1. **Force sensor**
   The pale blue area is Apple's own highlight on this frame. Press and swipe happen on that flat face. There is no mechanical button.
2. **Charging contact**
   The dark slot at the end of the stem. Charging only. It is not a microphone.
3. **Microphone mesh**
   The edge of the outward grille, at the left of this frame.



FIT

The tip, lifted

1. **Ear tip**
   Lifted off the nozzle. Foam-infused silicone, five sizes. The tip is rotated inward on this generation, which is Apple's fit change.
2. **Body mesh**
   An opening in the shell. Apple shows it in the fit photograph and does not give this mesh its own name, so this note does not invent one.
3. **Lower mesh**
   A second opening, at the lower end of the body in this view. Same rule: visible here, not separately named by Apple.



APPLE'S CUTAWAY

Inside the earbud

1. **Driver**
   The round speaker. Apple calls it a custom high-excursion driver. The diameter, impedance, and sensitivity are not published.
2. **H2**
   The chip in each earbud. Apple prints the name on this image. Listening modes, Adaptive EQ, and the Siri wake run with this chip.



OPEN

The case

1. **Status light**
   The green point on the front of the case. It is the status light. Pairing, on the reviews, is a tap on the case. The spec page does not describe a rear button.
2. **Left**
   The letter L in the left well. The left earbud sits here.
3. **Right**
   The letter R in the right well. Each earbud has its own microphones, so one bud is enough for a call.

How to read the figures

Blue is Active Noise Cancellation. Teal is Transparency. Amber is Adaptive Audio, and also the dashed passive rating where that rating is the reference. Stone is the foam seal with the electronics off. The same colour means the same thing on every figure. Figure 6 is a secondary citation, a set of category averages, not a frequency curve.

Three decibel tests are kept on separate figures. Apple's hearing-protection total is one number for a whole noisy room. Apple's octave table is the unpowered seal at one frequency. The SoundGuys curves are a single fixture sweep. A doubling of acoustic power is 3 dB. A sound that seems about half as loud is closer to 10 dB. Apple's "up to 2× more than AirPods Pro 2" is the power kind of claim, from an IEC 60268-24 test, and Apple does not publish that curve.

What active cancellation adds, by frequency

Blue fill is the gap between the seal alone and ANC on. Where the stone line rises above the blue, the foam is doing more than the electronics.

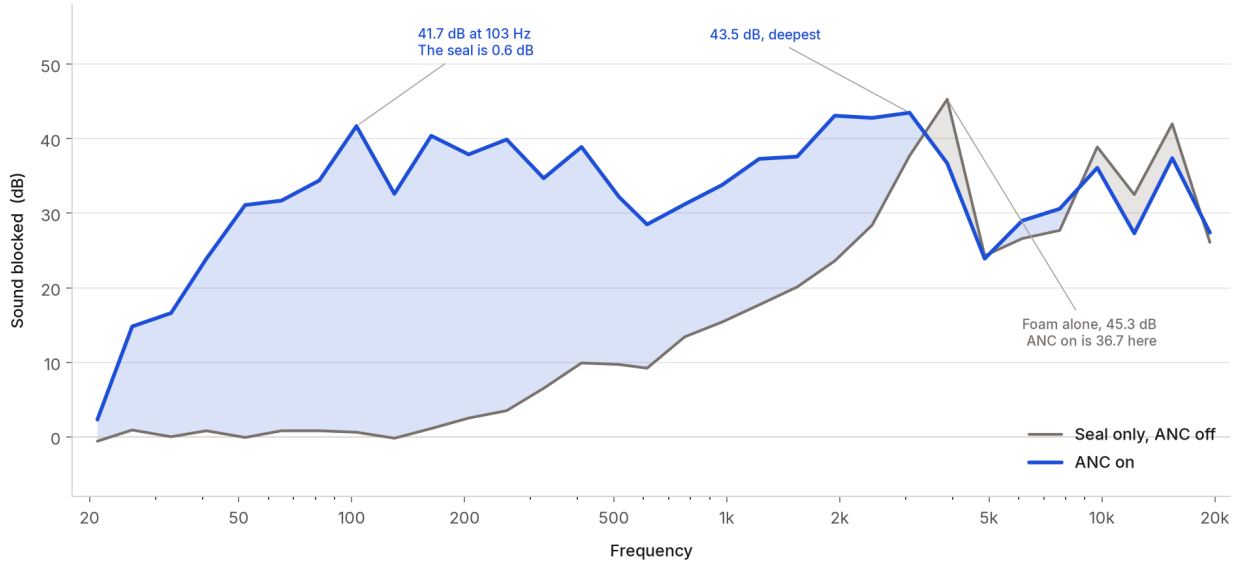


Figure 1 SoundGuys laboratory sweep on a B&K 5128, retrieved 5 November 2026. Not an Apple specification. One published sample.

Active cancellation is a low-frequency tool. Under about 200 Hz the foam blocks almost nothing, so the blue fill is nearly the whole of the protection. The deepest point on this sweep is 43.5 dB at 3.1 kHz. From about 4 kHz up, the seal is already strong, and at 3.9 kHz the foam alone measured 45.3 dB while ANC on measured 36.7 dB. Cancellation is not more blocking at every frequency.

Hearing protection as the room gets louder

The band is Apple's published range. The lower edge is what most people can expect. The upper edge is the proficient fit. Same vertical scale on every panel.

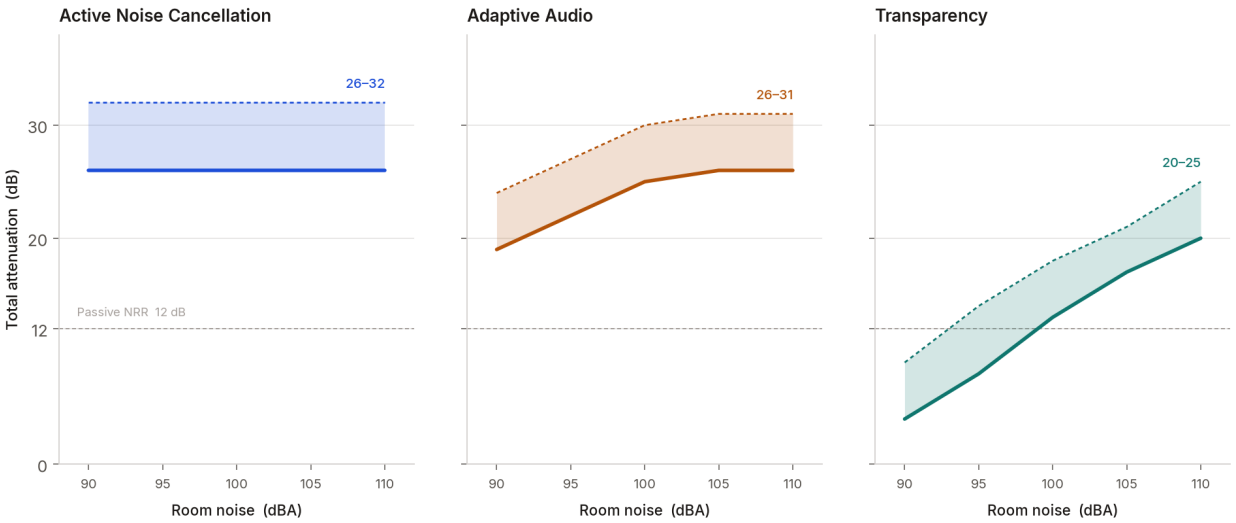


Figure 2 Apple Hearing Protection data sheet, AirPods Pro 3, September 2025. ANSI/ASA S12.68. Sustained noise only. Not for impulse sound, and not above 110 dBA.

ANC's estimated protection stays flat: 26 to 32 dB from a 90 dBA room through a 110 dBA room. Transparency starts near a light reduction, 4 to 9 dB at 90 dBA, and climbs to 20 to 25 dB at 110 dBA, because Loud Sound Reduction turns protection up as the room gets louder. Adaptive Audio sits between them and meets ANC once the room is very loud. The 12 dB dashed line is all that remains when the mode is Off or the battery is empty. These totals are for sustained noise. They are not for gunfire, fireworks, or jackhammers, and not for sound louder than 110 dBA.

The seal with the power off

Apple's only per-frequency decibel table. Real ears, battery removed. Whiskers are one standard deviation. The dashed line is the single-number NRR.

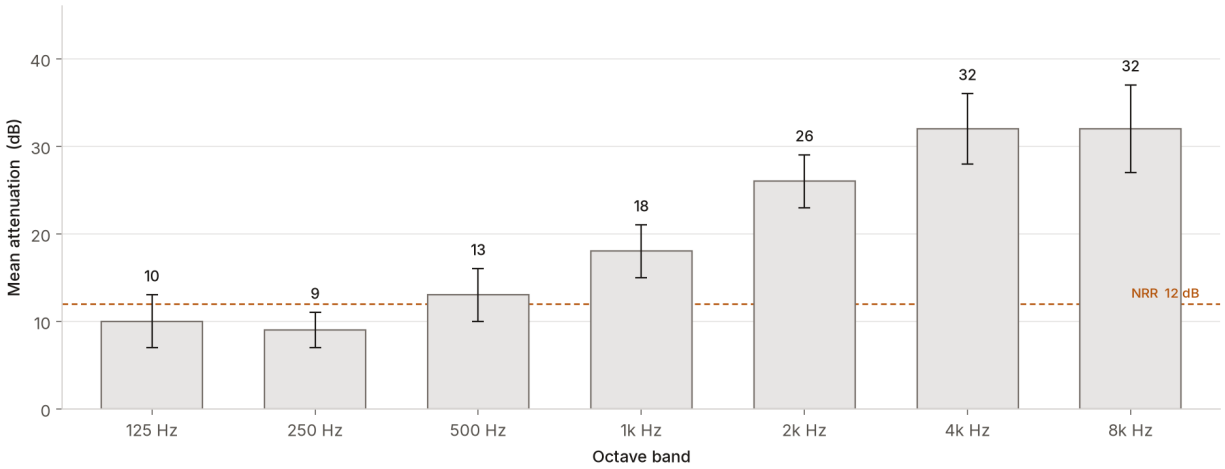


Figure 3 Apple, September 2025. ANSI S3.19-1974, unpowered. Standard deviations are 3, 2, 3, 3, 3, 4 and 5 dB. This is not the ANC curve.

With the battery out, a real-ear panel averaged about 10 dB at 125 Hz and 32 dB at 4 and 8 kHz. That shape is why ANC exists. The single-number rating of 12 dB is a summary of this uneven seal, not a promise of 12 dB at every frequency. Canada's CSA class for the passive performance is C.

Playback against a preference target

Zero is the SoundGuys target, not silence. Above the line is louder than that target. The three-band Custom EQ is a separate control, and Apple does not publish its

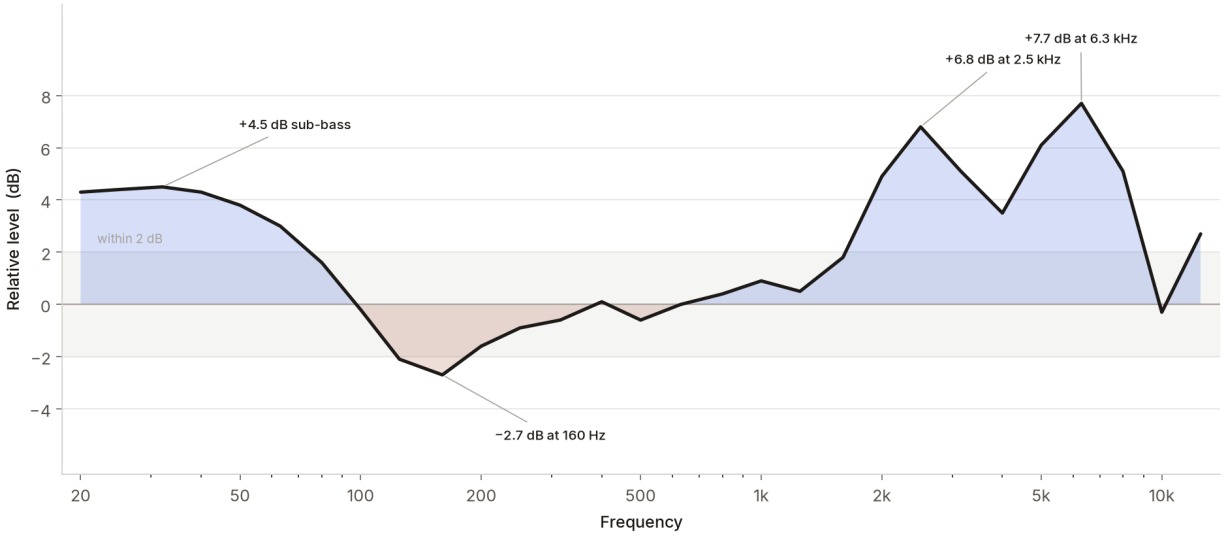


Figure 4 SoundGuys B&K 5128, relative to their preference curve. Off this scale: −11.5 dB at 16 kHz, −36.6 dB at 20 kHz.

On this laboratory comparison the sub-bass sits about 4.5 dB above the lab's target, 160 Hz sits 2.7 dB below it, and there are lifts of 6.8 dB at 2.5 kHz and 7.7 dB at 6.3 kHz. Adaptive EQ still retunes the driver from the inward microphone. Separately, iOS 27 adds three sliders, Low, Mid, and High, stored on the earbuds. Apple does not publish their decibel range, their center frequencies, or a Q. There is no 10-band equalizer on the product.

Hours on one charge

Apple's own test: 50% volume, 256 kbps AAC, preproduction units, July and August 2025. The case adds recharges, so 24 hours with the case is not drawn here.

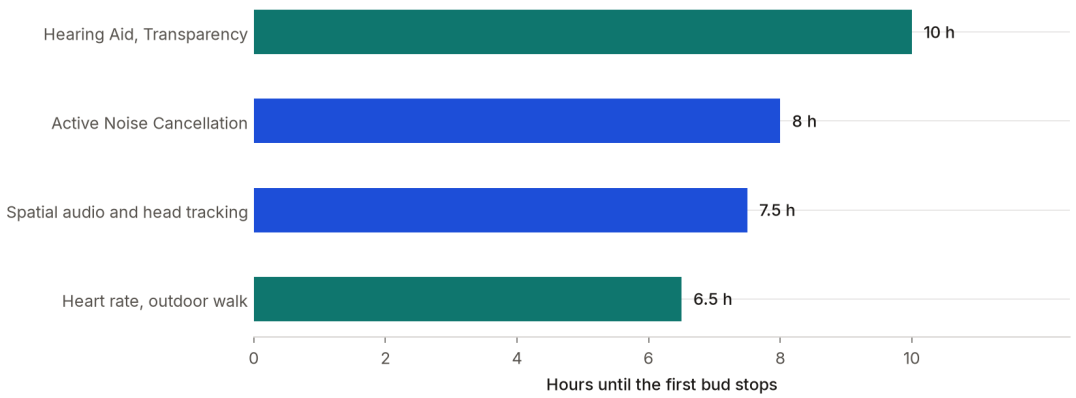


Figure 5 Apple, July–August 2025. Teal rows are Transparency. Blue rows are ANC. The 24-hour total with the case is off this scale.

Eight hours is the ANC figure at half volume. Spatial audio with head tracking is 7.5 hours. Heart-rate sensing during a walk is 6.5. Transparency with the Hearing Aid feature, and nothing playing from the phone, is 10. With the case, ANC listening is up to 24 hours, and five minutes in the case is about one hour. Putting 24 on the same axis would flatten the differences that matter on a single charge.

Category averages cited from RTINGS

These are single numbers for a band or a noise scene. They are not a frequency curve, and they are not Apple's hearing-protection test.

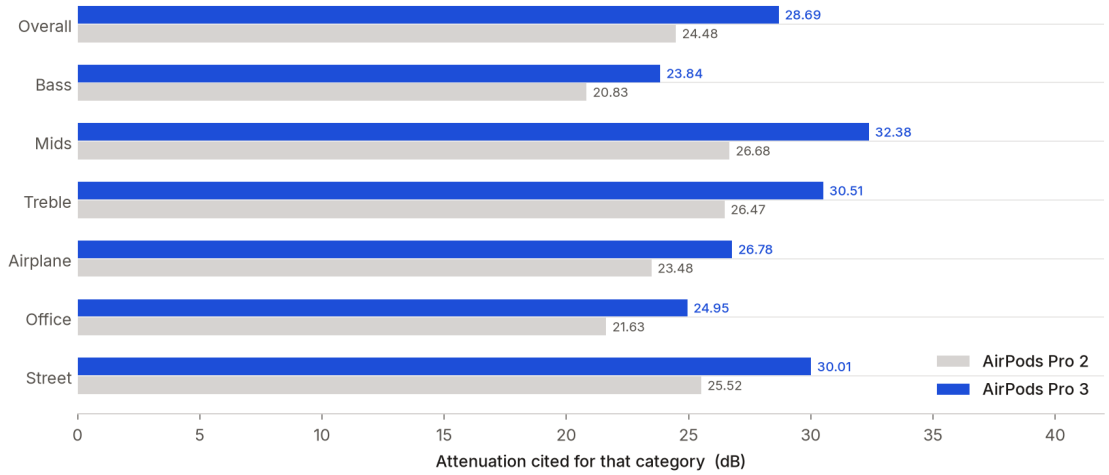


Figure 6 Public summary of an RTINGS review, October 2025. The review page is behind a login, so this figure is a secondary citation.

The step from 24.48 dB to 28.69 dB overall is 4.21 dB, about 2.6 times the acoustic power. That sits near Apple's "up to 2×" claim and is still a different test from the 26 to 32 dB hearing-protection band and from the SoundGuys sweep.

Identity, equalizer, and the case

Apple's identify page lists earbuds A3063, A3064, and A3065, and the MagSafe case A3122. A3047 and A3048 are AirPods Pro 2. The retail family is MFHP4. One India listing uses MFHP4HN/A. Apple's firmware list read for this revision shows 9A348. Build 8B30 was a December 2025 update.

iOS 27 adds a three-band equalizer: Low, Mid, and High, stored on the earbuds, under Audio and Routing, Equalizer, Custom. Recommended restores Apple's profile. Apple does not publish the slider range. Adaptive EQ, Headphone Accommodations, and the Hearing Aid audiogram remain separate controls. Charging is MagSafe, an Apple Watch charger, Qi, or USB-C. The spec page does not say Qi2. Reviews report that the rear pairing button is gone and that a tap on the case starts pairing.

The stem

There is no mechanical button. Each stem is a force sensor. A press is pressure on the stem, not a click. A light swipe on the same stem changes volume. Both stems can do every gesture. Press-and-hold is the one gesture that can be pointed at Siri instead of the listening modes.

Gesture	What it does	Reassigned?
Press once	Play or pause. Answer a call. Mute or unmute. Camera remote in a compatible app on iOS 26 or iPadOS 26.	Mute and end-call can swap with the double press.
Press twice	Next track. End a call, in the default setup.	End-call can swap with mute. Skip stays.
Press three times	Previous track.	No.
Press and hold	Cycle listening modes. The default cycle is ANC and Transparency.	Per stem: modes, or Siri. The cycle can add Adaptive Audio and Off.
Swipe up or down	Volume.	Not described as reassignable.
Hold both stems	Start Live Translation.	Fixed.
“Siri” or “Hey Siri”	Playback, calls, directions, and other Siri actions.	The hold can also call Siri.
Nod, or shake	Accept or decline an announced call or notification.	Only with Announce Calls and Announce Notifications on.

Automatic Ear Detection is on by default. Remove one earbud and playback pauses. Remove both and it stops, and does not resume by itself. With neither bud in the ear, audio comes from the phone. One bud in the ear is enough for music and for a call. Accessibility can turn on Noise Cancellation with One AirPod. Press Speed and Press and Hold Duration are in the same accessibility screen. The call microphone can be automatic, always left, or always right.

Six microphone channels

Apple lists two outward beamforming microphones and one inward microphone. Read once per earbud, the pair has six channels. The outward pair is shared: it aims at the mouth for calls, and it samples outside noise for feedforward cancellation. The inward microphone is shared too: it hears what the seal failed to stop, and it hears what the driver actually delivered so Adaptive EQ can correct the bass. They are not split into “two for ANC and four for voice.”

Sensor, each earbud	Count	Job
Outward beamforming microphones	2	Voice toward the mouth. Outside noise for feedforward ANC.
Inward microphone	1	Feedback ANC, and Adaptive EQ from the sound in the canal.
Speech-detecting accelerometer	1	Own-voice vibration. Not a microphone, and not speech-to-text.
Motion-detecting accelerometer	1	Head tracking, and the nod and shake gestures.
Skin-detect sensor	1	In-ear detection.
Heart-rate sensor	1	Optical pulse sensor for workouts on iOS 26. Not part of the audio path.

Speech, and what is actually decoded

The earbuds do not turn an arbitrary song, podcast, or meeting into a transcript. Where language is recognized, the paired iPhone does that work. When the earbuds speak, the words were written on the phone and played back.

Feature	What comes back	Where it runs
Siri	A spoken answer in the earbuds.	Wake can be recognized with the H2. The request is handled on the Apple device.
Announce Calls	The phone reads the caller aloud. A nod answers.	Text-to-speech on the phone.
Live Translation	Translated speech in the ear, and a live transcript of your words on the iPhone.	Apple Intelligence on a nearby iPhone, iOS 26 or later. Hold both stems. Beta.
Conversation Awareness	No text. Media drops and the mode opens when you start speaking.	On the earbuds, from the microphones and the speech accelerometer.
Hearing Aid	Louder speech from your audiogram. No transcript.	On the earbuds. FDA-authorized, age 18 or older, mild to moderate loss.
Live Listen	The iPhone microphone, played in the earbuds.	The phone is the remote microphone. No transcription.
Workout Buddy	Spoken encouragement, milestones, and a summary.	Apple Intelligence on the iPhone, iOS 27, English or Spanish.
Studio recording	The wearer’s voice, stored by a phone app.	Firmware notes name Camera, Voice Memos, and Messages dictation.

Live Translation languages on the Apple Store page retrieved 5 November 2026: Mandarin simplified and traditional, English of the UK and the US, French of France, German of Germany, Italian, Japanese, Korean, Portuguese of Brazil, and Spanish of Spain. At the September 2025 announcement the set was English, French, German, Portuguese, and Spanish.

What is not in this note

Driver diameter, impedance, sensitivity, distortion, latency in milliseconds, an official frequency-response curve, an official ANC curve against frequency, a Transparency frequency curve, and microphone part numbers. The three Custom EQ sliders exist, and Apple does not publish their decibel range, center frequencies, or Q. There is no 10-band equalizer. Apple lists studio-quality recording and does not publish the sample rate or the bit depth. A later firmware can move a laboratory curve. The opening photographs and the numbered part plates are Apple’s, from apple.com/airpods-pro and the Apple Store. A mesh that Apple does not name is not given a name here.

Sources

Apple technical specifications, apple.com/airpods-pro/specs and support article 125135. Apple Newsroom, 9 September 2025. Identify your AirPods, support article 109525. About firmware updates, retrieved 5 November 2026, listing 9A348. Customize EQ on AirPods, and settings article 108764, 14 September 2026. Accessibility guide for AirPods. Workout Buddy, Apple Fitness guide, iOS 27. Apple Store language list, retrieved 5 November 2026. Hearing Protection data sheet, September 2025, regulatory file A3063 / A3064, and support article 120850. SoundGuys laboratory tables, retrieved 5 November 2026. RTINGS category averages are a secondary citation from a public summary of the review, October 2025. The review page itself is behind a login.